CSD 340 practice problems for LMM - HW6

1. Consider the R formulas for the following two LMM models: given that X_1 is quantitative and X_2, X_3 standards for the group IDs from 2 different sampling way of the observations,

M1: $Y \sim X_1 + (1|X_2) + (1|X_3)$

M2: $Y \sim X_1 + (X_1|X_2)$

a) List each parameter of M1 and give the corresponding interpretation.
b) Given the estimated parameters and EBLUPs for M1, find the mean and sd for the unconditional Y, also for the conditional Y for M1.
c) List each parameter of M2 and give the corresponding interpretation.
d) Given the estimated parameters and EBLUPs for M2, find the mean and sd for the unconditional Y, also for the conditional Y for M2.

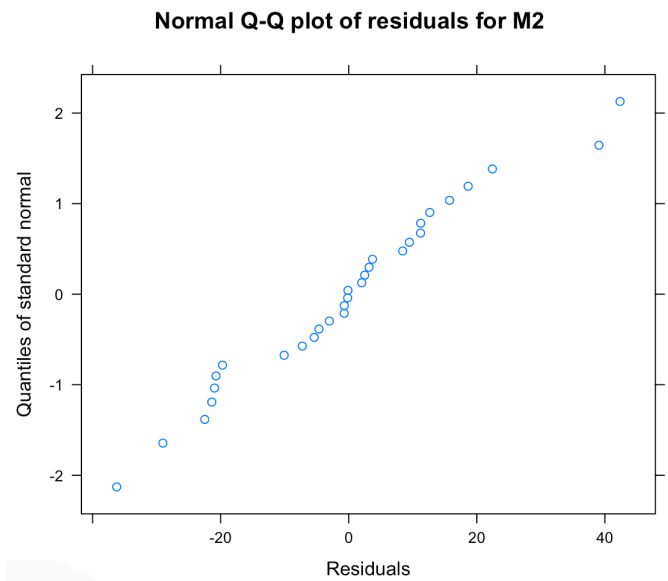
2.Consider the data in #3 of HW4 and model designed for it in #4 in HW5. After making "region" and "treatment" into factor features, three models are fitted on the data and their information is given in the table below.

Model	Fixed factors	Random factors	Variances of residuals R_i	logML	logREML
M0	Region, treatment	Intercept	$\sigma^2 I_6$	-175.4605	-158.1243
M1	Region, treatment, region*treatment	Intercept		-162.6688	-137.6411
M2		Intercept, treatment		-146.3615	-124.5952
M3			σ^2_{treat} for treatment, $\sigma^2_{control}$ for control.	-146.2664	-124.5192

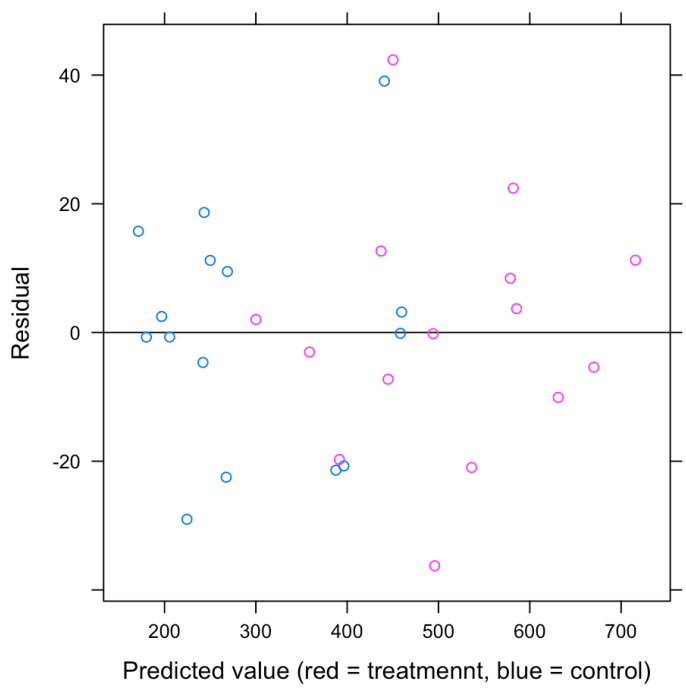
a) Is the random effects associated with treatment for each animal significant? Perform a likelihood ratio test to determine.
b) For the residual error variance, which one is more appropriate: homogeneous σ^2 or different σ^2_{treat} and $\sigma^2_{control}$ for treatment and control group respectively? Perform a likelihood ratio test to determine.
c) Is the fixed effects region*treatment significant? Perform a likelihood ratio test to determine.

d) Among the four models, which one should be chosen to be the final model?

e) The distribution of conditional residuals from the final model is shown as below. Comment on it about the model's performance.



f) The scatterplot of conditional residuals vs conditional predicted values based on the fit of the final model is given below. Comment on it about the model's performance.



g) Below are M2's model summary (part of it) and EBLUPs displayed in R. Answer the question following.

Part of M2's model summary:

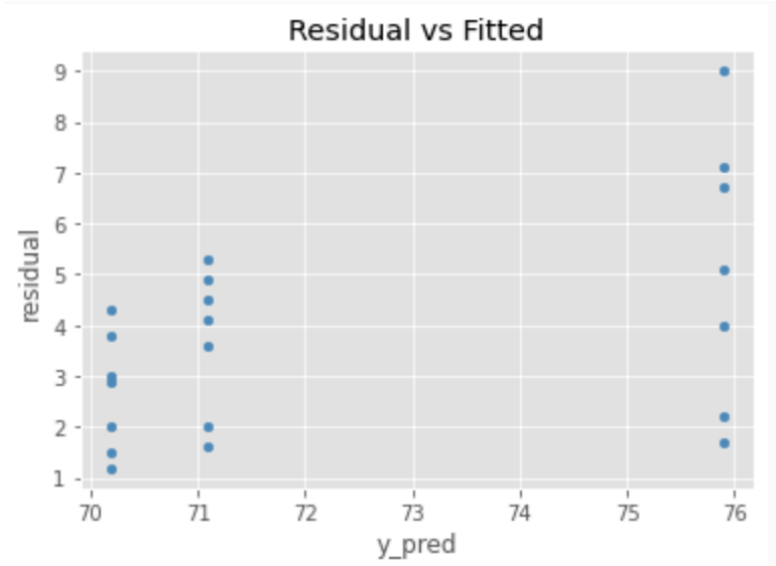
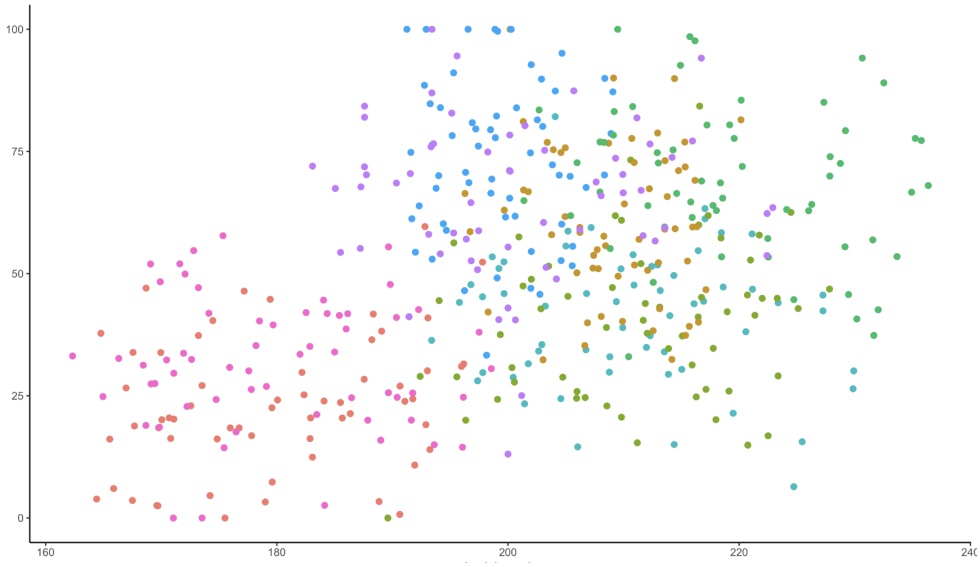
```
Linear mixed-effects model fit by REML
      AIC      BIC    logLik
  269.1904 280.971 -124.5952

Random effects:
Formula: ~treatment | animal ID
Structure: General positive-definite, Log-Cholesky parametrization
                StdDev   Corr
(Intercept) 35.83696 (Intr)
treatment   79.82012 0.801
Residual    23.21432

Fixed effects:  activate ~ region * treatment
                          Value Std.Error DF    t-value p-value
(Intercept)               212.294  19.09551 20  11.117483  0.0000
region1                   216.212  14.68203 20  14.726304  0.0000
region2                   25.450   14.68203 20   1.733412  0.0984
treatment                 360.026  38.59808 20   9.327561  0.0000
region1:treatment    -261.822  20.76352 20 -12.609710  0.0000
region2:treatment    -162.500  20.76352 20  -7.826225  0.0000
```

EBLUPs of M2:

```
       (Intercept)     treatment
R3    29.77622    113.624834
R4    12.23877     -2.696041
R5    31.10922     27.725298
R1   -40.92650     -94.351747
R2   -32.19771     -44.302345
```

1. What is the estimated random effect covariance matrix D and the residual covariance matrix R_{ij}.
2. What is the conditional fitted activation value for the observation with observed activation 587.10? What is the unconditional fitted activation value?
3. Write down the LMM equation to model the response assumed by M2 with the estimated parameters given above.

3. Comment on the model with following residual vs fitted y plot.



4. Given the y_fitted vs residual plot of the model M0, where different color standards for different country's data points, comment on M0 and provide another model M1 which may outperform M0.